

Pain Is Subjective, Reassessment Is Measurable: Understanding Variation in Emergency Department Pain Reassessment

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Abstract

Background: Pain is the most common presenting symptom in the emergency department, yet its subjective nature makes it uniquely vulnerable to inequities in clinical response. Disparities in analgesic treatment are well-documented, but pain reassessment — whether a clinician returns to check on a patient after an initial pain score is recorded — is an upstream, quantifiable process-of-care measure that shapes treatment decisions and patient outcomes yet remains poorly characterized.

Methods: We conducted a retrospective cohort study of emergency department (ED) encounters with documented pain assessments using the MIMIC-IV Emergency Department database. The primary outcome was time from the initial documented pain score to the first subsequent reassessment, analyzed using Cox proportional hazards models with sequential adjustment for clinical presentation (M1), demographic and social factors (M2), illness severity and triage (M3), and ED workflow context (M4). Sensitivity analyses included within-acuity models, severe-pain subsets, post-analgesic reassessment timing, and year-over-year trend analyses.

Results: Among 17,204 encounters, 10,569 (61.4%) had a documented reassessment during the ED stay. Median time to reassessment was 185 minutes (IQR 96–311); only 9.6% of patients were reassessed within 60 minutes. Higher initial pain score (HR 1.068 (95% CI 1.060–1.076), $p < 0.001$) and ambulance arrival (HR 1.097 (95% CI 1.050–1.146), $p < 0.001$) were associated with faster reassessment. Medicaid insurance was associated with substantially slower reassessment compared with private insurance (HR 0.807 (95% CI 0.763–0.853), $p < 0.001$), with minimal attenuation across sequential models and within triage-acuity strata — including among patients with the highest-acuity triage designations and among those reporting severe pain. Race and ethnicity associations in unadjusted models attenuated substantially after adjustment for ESI triage acuity. Critically, this attenuation is consistent with ESI acting as a mediator of racial disparities in reassessment — reflecting differential triage assignment for equivalent presentations — rather than evidence that no disparity exists. Reassessment rates did not improve meaningfully across the study period (HR per 5-year increase 0.999 (95% CI 0.996–1.003, $p = 0.793$)).

Conclusions: Pain reassessment is frequently delayed and inequitably distributed. Insurance-based differences persist after full adjustment, pointing to unmeasured factors in the care encounter. Race/ethnicity effects appear to operate through triage pathways, implicating acuity assignment itself as a mechanism of inequity. Because reassessment is directly observable and protocol-adjacent, both findings identify concrete targets for quality improvement.

1 Introduction

Pain is among the most common reasons patients seek emergency care, yet it remains one of the most challenging clinical symptoms to assess and manage. Unlike objective physiologic measurements,

pain is inherently subjective and relies entirely on patient self-report, making it vulnerable to variation in interpretation, documentation, and clinical response. Because pain cannot be directly observed or verified from the outside, its evaluation depends on a structured chain: a patient reports their pain, a clinician documents that report, and subsequent reassessments determine whether the symptom has changed and whether the response to treatment has been adequate. Each step in this chain introduces an opportunity for variation — and potentially for inequity.

Pain reassessment is a critical component of high-quality emergency care. Reassessment provides an opportunity to determine whether symptoms have improved, whether additional treatment is warranted, and whether the patient’s clinical trajectory is changing. Failure to reassess pain may contribute to undertreatment, delayed recognition of persistent or worsening symptoms, and inconsistent patient experiences across the ED visit. Importantly, because reassessment begins with the documentation of a patient’s reported pain, variation in reassessment practices represents an early and measurable point at which differences in care quality emerge.

From a quality-improvement perspective, reassessment is attractive precisely because it is observable. Unlike the underlying experience of pain — which cannot be extracted from the medical record — whether and when a clinician returns to document a follow-up pain score is a concrete, auditable event. This distinguishes reassessment from many other potential disparity targets: it is protocol-adjacent, not dependent on a subjective clinical judgment, and therefore directly amenable to structured intervention. That we know so little about who receives timely reassessment and who does not represents a meaningful gap, particularly given the well-documented disparities in pain treatment at the point of analgesic administration.

Prior studies have largely focused on analgesic administration, opioid prescribing, and treatment disparities. Pain reassessment is upstream of and distinct from these outcomes: it reflects whether the clinical system is monitoring a patient at all, independent of what treatment decisions follow. Disparities in reassessment may therefore reveal inequities in monitoring intensity that would be invisible in treatment-centered analyses — and may persist even in settings where analgesic disparities have been addressed.

Reassessment timing is shaped by multiple overlapping factors. Clinical characteristics such as pain severity, diagnosis, and physiologic instability may appropriately drive faster reassessment. At the same time, triage acuity, operational ED conditions, arrival characteristics, and patient demographics may also influence how quickly the system responds to a patient’s reported pain. Critically, triage acuity may lie on the causal pathway between patient identity and reassessment, rather than acting as an independent confounder. If disparities in reassessment operate partly through the way triage encodes clinical urgency — that is, if equivalent presentations receive different acuity assignments depending on patient characteristics — then standard adjustment for ESI removes disparity from the model without removing it from the data. This distinction has direct implications for where quality-improvement efforts should be targeted.

The objective of this study was to characterize patient, clinical, and operational factors associated with time to first pain reassessment in the emergency department, using a sequential model-

ing framework designed to distinguish early clinical signals from downstream triage and workflow context. Rather than focusing on a single demographic characteristic, we examined the broader structure of factors that predict system responsiveness to a patient’s own report of their pain.

2 Methods

2.1 Study design and data source

We conducted a retrospective cohort study of ED encounters with documented pain assessments using the Medical Information Mart for Intensive Care IV (MIMIC-IV) Emergency Department module. MIMIC-IV contains de-identified clinical data from a single tertiary academic medical center; calendar years are reported as de-identified anchor years to prevent re-identification. The study was designed to characterize factors associated with time to first pain reassessment across routine ED care, with particular attention to the causal structure linking patient characteristics, triage, and operational context to reassessment timing. This report follows STROBE guidance for observational cohort studies [von Elm et al., 2007].

2.2 Conceptual framework and causal structure

Pain reassessment occupies a specific position in the care pathway: it follows both an initial patient-reported pain score and a triage evaluation, and precedes definitive disposition. To clarify the relationships among covariates and to motivate the sequential modeling strategy, we specified a directed acyclic graph (DAG) prior to analysis (Figure 1).

The DAG encodes three key structural features. First, initial pain score and diagnosis are upstream clinical signals that both trigger reassessment directly and inform triage acuity; they represent the *clinical presentation* tier (M1). Second, patient identity variables — insurance status, race/ethnicity, and preferred language — may influence reassessment through two distinct pathways: an indirect pathway operating through triage acuity (if patient identity shapes how clinical urgency is assigned) and a residual direct pathway not mediated by triage. Third, ESI triage acuity occupies an ambiguous structural position: it may act as a confounder (if it reflects true underlying severity that independently predicts reassessment), a mediator (if the same clinical presentation receives different acuity assignments depending on patient identity), or both. This ambiguity motivates the sequential modeling strategy described below, and is the reason we report results from models that progressively add covariate tiers rather than selecting a single adjusted specification.

ED operational factors — arrival mode, shift timing, and crowding — are downstream of both triage and patient identity. Ambulance arrival in particular may lie on the pathway from insurance to reassessment (Medicaid patients are less likely to arrive by ambulance), and is therefore included in M4 as an operational covariate whose effect should be interpreted as part of the care pathway rather than an independent confounder to be removed.

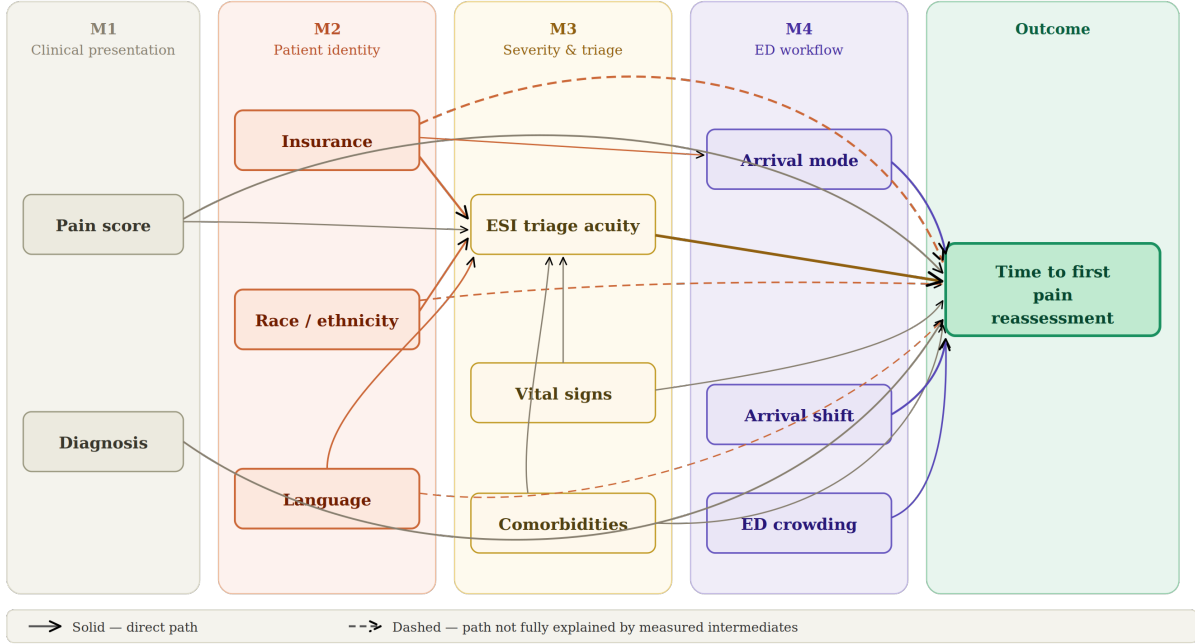


Figure 1: Directed acyclic graph (DAG) of hypothesized causal relationships between patient, clinical, and operational factors and time to first pain reassessment. Nodes are organized into four tiers corresponding to the sequential covariate domains (M1–M4). Solid arrows represent hypothesized direct causal paths; dashed arrows represent paths whose magnitude is not fully explained by measured intermediates. Patient identity variables (M2) may influence reassessment both through triage acuity — if equivalent presentations receive different ESI assignments depending on patient characteristics — and through pathways that bypass triage entirely. The sequential modeling strategy is designed to distinguish these two routes: attenuation of a demographic coefficient after adding ESI (M2 to M3) is interpreted as evidence that the effect operates through differential triage assignment, while persistence across all models points to unmeasured mechanisms in the care encounter.

2.3 Study population

We identified ED stays with at least one documented initial pain score and eligible ED diagnoses. The analytic cohort was restricted to encounters with diagnoses of acute pancreatitis or trauma (including fall, fracture/dislocation, and other trauma subtypes). This restriction served a deliberate design purpose: both conditions carry a consistent clinical indication for ongoing pain monitoring, reducing heterogeneity in the expected reassessment standard and limiting the influence of diagnosis-specific documentation practices on the outcome. Cohort construction is summarized in Figure 2.

Starting from 47,783 ED stays in the extract, we applied diagnosis-based inclusion, excluded small or unknown race/ethnicity categories, required a positive initial pain score with valid time-to-event information, excluded undocumented insurance, and required a documented ESI triage level. The resulting primary analytic cohort comprised 17,204 encounters.

A note on missing demographic data: encounters with undocumented insurance were excluded from the primary cohort because insurance category is a primary exposure variable and missingness in this field likely reflects systematic documentation differences rather than random omission. We treat this exclusion as a study design decision rather than a missing-data problem, and report the volume of excluded encounters in Figure 2. Sensitivity analyses among a cohort with complete demographic documentation are reported in the supplement.

2.4 Outcome and follow-up

The primary outcome was time from the **initial documented pain score** to the **first subsequent pain reassessment**, measured in minutes. This framing operationalizes the key research question: given that a patient’s pain has been documented, how quickly does the clinical system respond with a follow-up assessment? Patients without a documented reassessment before ED departure were censored at the end of the ED stay. A reassessment was defined as any pain score recorded strictly after the initial pain documentation timestamp.

For descriptive reporting, we also calculated the proportion of patients reassessed within 60 and 120 minutes of the initial documentation. In sensitivity analyses, we examined post-analgesic reassessment timing with time zero set to the first analgesic administration after initial pain documentation (Section 2.6).

2.5 Covariates and sequential model design

Covariates were organized into four sequential tiers, each corresponding to a distinct phase of the care pathway and a distinct causal tier in Figure 1:

- **M1 (clinical presentation):** initial pain score (continuous) and injury/diagnosis group (acute pancreatitis as reference category). These variables represent the clinical signal that initiates the reassessment process — the patient’s own report of their pain severity, and the clinical context in which it arises.

Fig. 2. Cohort construction and analytic sample

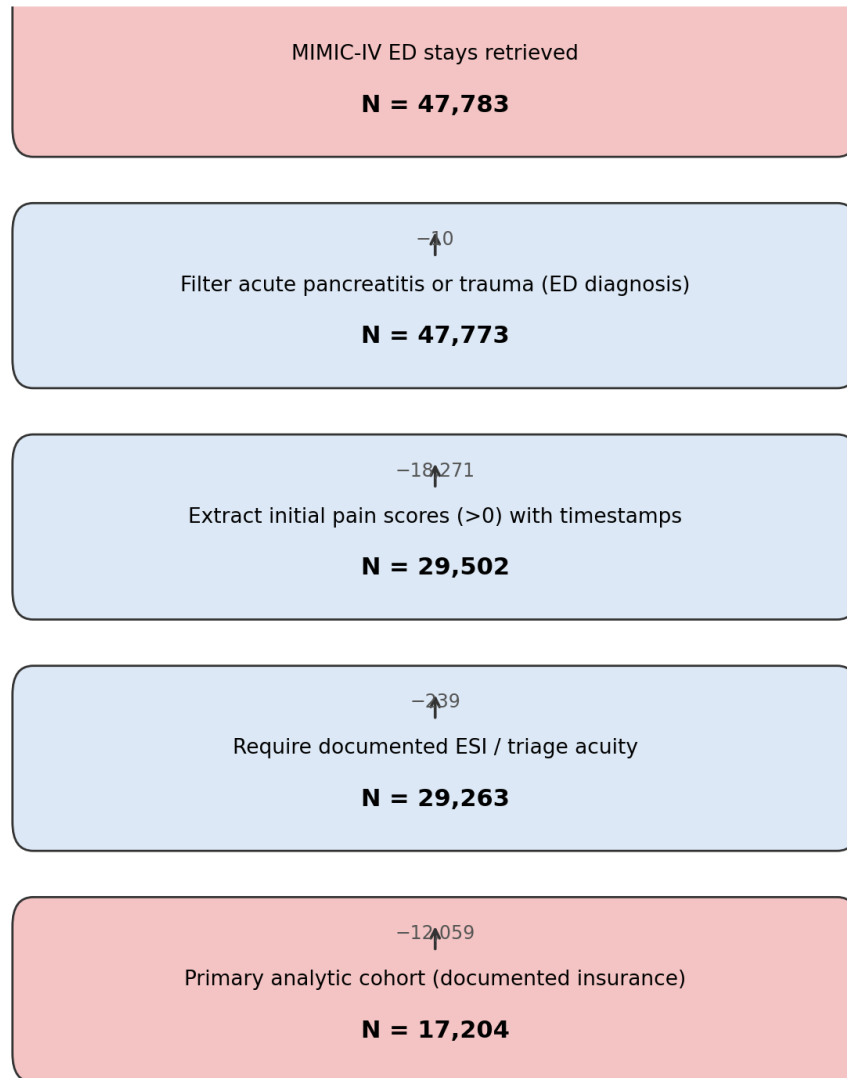


Figure 2: Cohort construction for the primary analytic sample. Starting from 47,783 ED stays in the MIMIC-IV extract, sequential exclusions yield the primary analytic cohort of 17,204 encounters.

- **M2 (demographic and social factors):** race/ethnicity (White as reference), insurance (private as reference), preferred language (English as reference when available), age group, and sex. These variables are upstream of triage assignment in the DAG and represent patient characteristics that may shape how the clinical signal is received and acted upon, independent of its objective severity.
- **M3 (illness severity and triage):** ESI triage level, initial vital signs (heart rate, respiratory rate, and systolic blood pressure, each z-scored), and comorbidity count derived from linked ICD-based flags. As described in the conceptual framework, ESI occupies an ambiguous causal position. We include it at this tier to quantify how much of the demographic associations in M2 operate through triage assignment, while acknowledging that this adjustment may partially block causal pathways of interest in addition to controlling for confounding.
- **M4 (ED workflow context):** arrival mode (walk-in as reference), arrival shift (day shift as reference), weekend arrival, and ED crowding proxies — the number of new arrivals in the hour preceding the initial pain documentation and the total census at that hour. These variables capture operational conditions that vary across visits independently of patient characteristics and that may modify how quickly any given patient is monitored.

Two variable categories were intentionally excluded from the primary sequential models. Disposition (admitted vs. discharged) occurs downstream of initial pain documentation and treatment decisions; including it would induce collider bias on the pathway between presentation and reassessment. Analgesic treatment similarly falls between initial pain assessment and reassessment and is treated as a pathway variable in dedicated sensitivity analyses rather than a confounder. Vital-sign coefficients are estimated in M3 and reported in the supplement; they are included for confounding control but are not the focus of primary inference.

2.6 Statistical analysis

We used Cox proportional hazards models to estimate hazard ratios (HRs) for time to first pain reassessment. An HR greater than 1 indicates faster reassessment (shorter time to event); an HR less than 1 indicates slower reassessment. We fit four models (M1 through M4) adding covariate tiers sequentially in the order above, with **M4** prespecified as the primary adjusted model for inference on demographic, insurance, and workflow associations.

The sequential structure serves an interpretive purpose beyond simple covariate adjustment. The change in a coefficient from M2 to M3 quantifies how much of the demographic association is attributable to differential triage assignment — that is, how much of the apparent effect of race or insurance on reassessment operates through ESI acuity. The change from M3 to M4 quantifies additional attenuation through operational workflow factors. Coefficients that remain stable across M3 and M4 — as observed for insurance — indicate associations that are not explained by triage mix or operational context and may reflect structural or unmeasured factors in the care encounter.

Sequential attenuation was examined graphically for the three primary exposure variables (race/ethnicity, insurance, and initial pain score) across all four models (Figure 6).

Additional analyses.

1. **Within-acuity models:** M4-specification models fit separately within ESI strata (1–2, 3, and 4–5), to evaluate whether demographic associations persist after removing cross-stratum ESI variation entirely. This analysis addresses whether the insurance association reflects triage-level confounding not adequately controlled by including ESI as a continuous covariate.
2. **Severe-pain sensitivity:** Primary analysis restricted to encounters with initial pain scores of 7–10 (and a secondary restriction to pain score 10 only), to evaluate whether findings hold among patients with unambiguously high self-reported pain burden.
3. **Post-analgesic pathway analysis:** Time zero reset to the first analgesic administration after initial pain documentation, analyzing reassessment conditional on having received treatment. This analysis separates whether the system responds to a patient’s initial pain report (primary outcome) from whether it monitors the response to treatment (post-analgesic outcome).
4. **Year-over-year trend analysis:** Unadjusted and M4-adjusted probabilities of reassessment within 60 minutes compared across de-identified 5-year policy eras, with a supplementary continuous-year Cox model. This analysis evaluates whether reassessment documentation practices changed across the observation window.
5. **Disposition pathway sensitivity:** M4 compared with M4 plus admitted-vs-home disposition as a covariate. Because disposition is a post-treatment variable, this model is not preferred for primary inference; it is reported to assess the degree to which disposition mediates workflow associations.
6. **Inverse probability of treatment weighting (IPTW):** Medicaid versus private insurance comparison with propensity scores estimated using M1 through M3 covariates only (excluding workflow), as a complementary approach to assess robustness of the primary regression-based estimate.

Analyses were conducted in Python 3 using `lifelines` and `statsmodels`; code is available in the study repository (`scripts/run_analysis.py`).

2.7 Ethics

MIMIC-IV is a publicly available, de-identified database; credentialed users complete required training and data use agreements. Institutional review board approval was not required for secondary analysis of de-identified data per the data use agreement.

3 Results

3.1 Cohort characteristics and overall reassessment patterns

The analytic cohort included 17,204 encounters; 10,569 (61.4%) had at least one documented pain reassessment during the stay. Cohort characteristics are shown in Table 1. The mean age was 56.2 years (SD 19.7); 70.4% of patients were White, 19.7% Black, 6.9% Hispanic, and 3.1% Asian. Most encounters were trauma-related (92.8%); acute pancreatitis accounted for 7.2%. Insurance distribution was Medicare 46.6%, private 29.5%, and Medicaid 23.9%. Mean initial pain score was 6.1 (SD 2.6); mean ESI was 2.7 (SD 0.7). Over half of patients arrived by ambulance (51.4%); 8.0% had a non-English preferred language.

Table 1: Characteristics of the analytic cohort ($N = 17,204$)

Characteristic	Value
<i>Demographics</i>	
Age (years), mean (SD)	56.2 (19.7)
Female sex, n (%)	9,797 (56.9)
White race/ethnicity, n (%)	12,108 (70.4)
Black race/ethnicity, n (%)	3,383 (19.7)
Hispanic ethnicity, n (%)	1,186 (6.9)
Asian race/ethnicity, n (%)	527 (3.1)
<i>Insurance</i>	
Medicare, n (%)	8,015 (46.6)
Private, n (%)	5,071 (29.5)
Medicaid, n (%)	4,118 (23.9)
Non-English preferred language, n (%)	1,380 (8.0)
<i>Clinical</i>	
ESI triage level, mean (SD)	2.7 (0.7)
Initial pain score, mean (SD)	6.1 (2.6)
Trauma diagnosis, n (%)	15,958 (92.8)
Acute pancreatitis, n (%)	1,246 (7.2)
<i>ED workflow</i>	
Ambulance arrival, n (%)	8,845 (51.4)
Weekend arrival, n (%)	4,907 (28.5)
Night shift arrival, n (%)	2,980 (17.3)
ED arrivals in prior hour, mean (SD)	2.1 (1.3)
<i>Outcomes</i>	
Any reassessment, n (%)	10,569 (61.4)
Reassessed within 60 min, n (%)	1,651 (9.6)
Reassessed within 120 min, n (%)	3,722 (21.6)
Time to reassessment, median (IQR)	185 (96–311) min

Reassessment was frequently delayed. The median time to first reassessment was 185 minutes (IQR 96–311). Only 9.6% of patients were reassessed within 60 minutes and 21.6% within 120 minutes, suggesting that timely pain monitoring is the exception rather than the norm in this cohort. Unadjusted Kaplan–Meier curves showed separation by insurance and race/ethnicity (Figure 3); 60-minute reassessment rates varied by pain severity stratum (Figure 4).

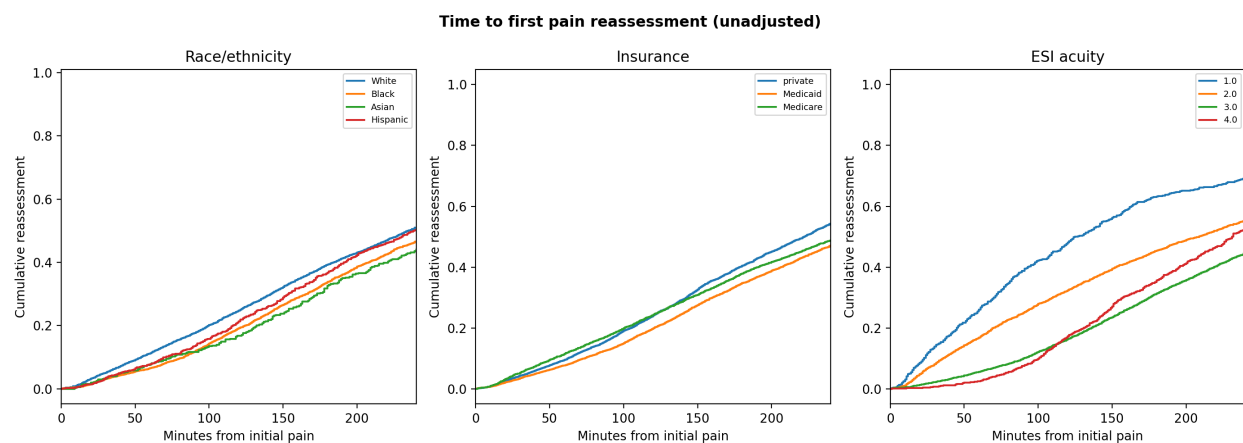


Figure 3: Unadjusted cumulative probability of first pain reassessment by race/ethnicity, insurance, and ESI triage level.

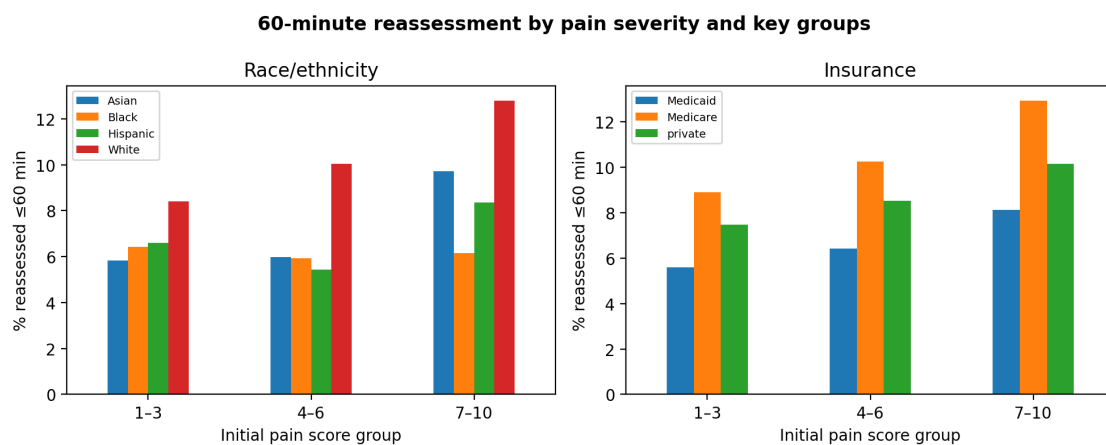


Figure 4: Unadjusted proportion reassessed within 60 minutes by initial pain severity group and race/ethnicity or insurance.

3.2 Clinical presentation: the primary driver of reassessment

In the primary adjusted M4 model (Figure 5), clinical presentation variables showed the strongest and most clinically interpretable associations with reassessment timing. Each one-point increase in initial pain score was associated with a meaningfully faster rate of reassessment (HR 1.068 (95% CI 1.060–1.076), $p < 0.001$): patients who reported higher pain were more likely to be checked on sooner, consistent with appropriate clinical prioritization of a patient’s own symptom report.

Higher ESI level (lower triage acuity) was associated with substantially slower reassessment (HR 0.830 (95% CI 0.805–0.856), $p < 0.001$). This association reflects the central role of triage in organizing clinical attention: patients assigned lower acuity receive less intensive monitoring, and this effect is large relative to most other covariates in the model. As described in the Methods, ESI is interpreted here as both a clinically appropriate driver of monitoring intensity *and* a potential pathway through which patient identity influences reassessment (see Section 3.4 and Figure 1).

3.3 Operational workflow factors

Ambulance arrival was associated with faster reassessment compared with walk-in patients (HR 1.097 (95% CI 1.050–1.146), $p < 0.001$), consistent with higher-acuity routing and more intensive initial evaluation for patients arriving by EMS. Night shift was also associated with faster reassessment (HR 1.100 (95% CI 1.039–1.164), $p = 0.001$), potentially reflecting lower census and staffing patterns that allow more frequent patient contact relative to the day shift. Weekend arrival and ED crowding variables showed weaker or null associations at conventional significance thresholds, suggesting that short-term operational variation in volume may not substantially alter reassessment behavior within this cohort.

3.4 Demographic pathways: race, ethnicity, and the role of triage

Race and ethnicity showed associations with reassessment timing in earlier models that attenuated substantially with sequential adjustment (Figure 6). In M2 (demographic adjustment only), Black patients had slower reassessment than White patients (HR 0.891 (95% CI 0.846–0.939), $p < 0.001$). This association attenuated toward the null after adding ESI and severity covariates in M3, and remained modest in the fully adjusted M4 model (HR 0.942 (95% CI 0.894–0.994), $p = 0.028$). Hispanic patients showed no meaningful association in any model (HR 1.035 (95% CI 0.951–1.128), $p = 0.425$). Asian patients had slower reassessment in M4 (HR 0.876 (95% CI 0.774–0.991), $p = 0.035$), but confidence intervals widened in severe-pain sensitivity analyses, limiting the precision of this estimate.

This pattern of attenuation carries a substantive interpretation rather than simply indicating that demographic effects disappear with adjustment. The DAG (Figure 1) encodes the hypothesis that patient identity variables may influence triage acuity assignment: if identical clinical presentations receive different ESI levels depending on who the patient is, then ESI is a mediator of the race/reassessment relationship rather than a confounder to be removed. Under this interpretation,

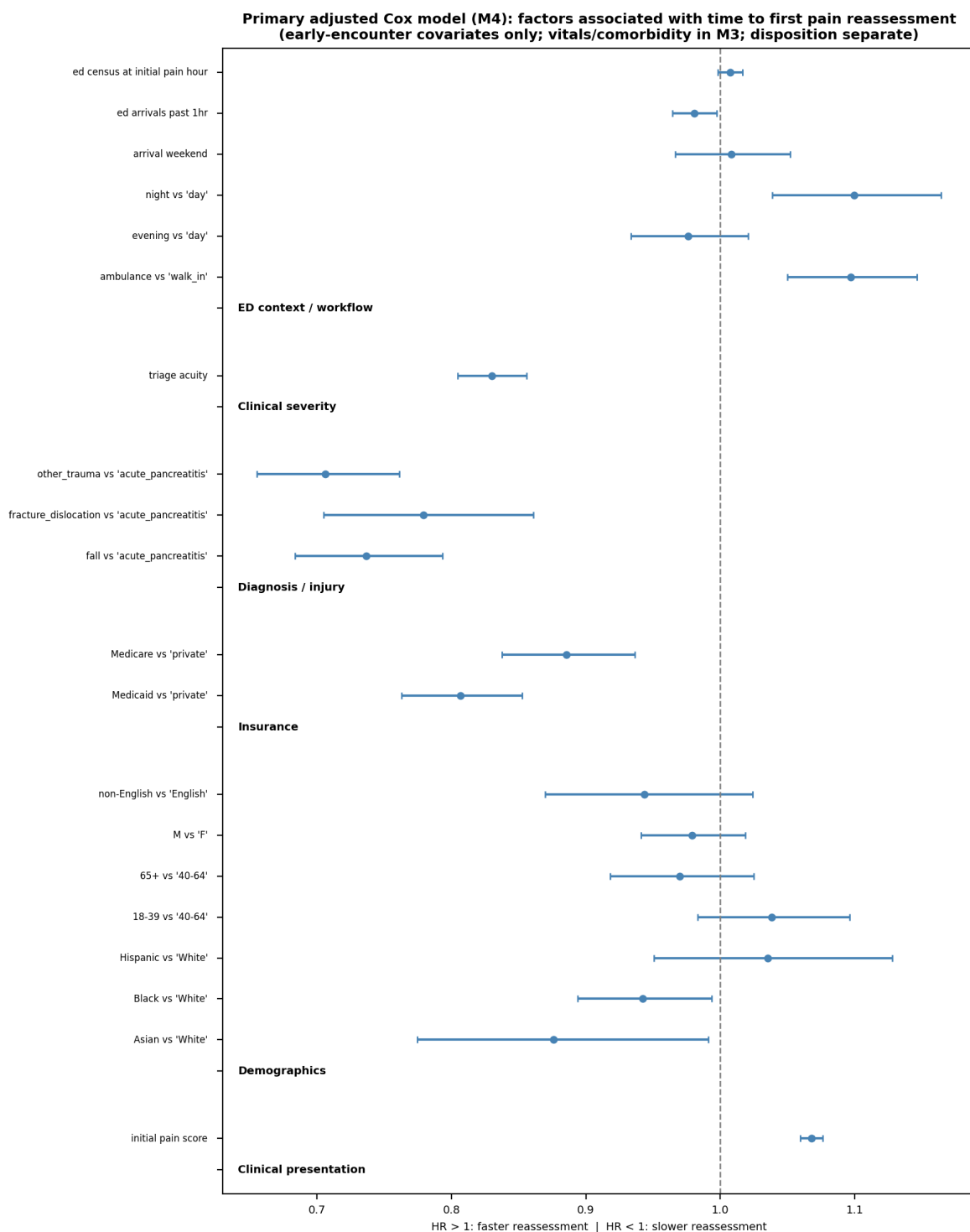


Figure 5: Primary adjusted Cox model (M4): hazard ratios for time to first pain reassessment. HR > 1: faster reassessment; HR < 1: slower reassessment. Vitals are adjusted in M3 but reported in the supplement; other arrival mode is excluded from the main panel.

adjusting for ESI partially blocks the causal pathway of interest, and the residual M4 coefficient should be understood as the portion of the race effect *not* mediated by differential triage assignment. That the Black–White coefficient attenuates substantially but does not fully resolve suggests that ESI assignment accounts for a meaningful share of the demographic disparity in reassessment.

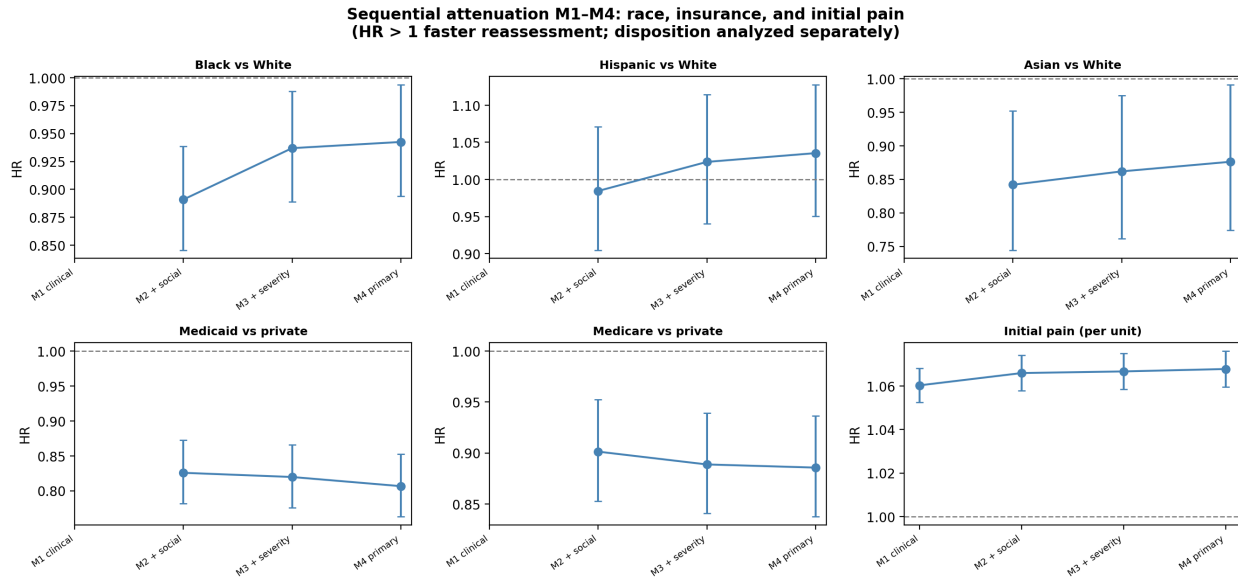


Figure 6: Sequential attenuation of race/ethnicity, insurance, and initial pain across models M1–M4.

3.5 Insurance: a persistent association not explained by clinical or triage factors

Insurance showed the largest and most robust associations with reassessment timing among the social and demographic variables. Compared with private insurance, Medicaid was associated with substantially slower reassessment in the fully adjusted model (HR 0.807 (95% CI 0.763–0.853), $p < 0.001$), representing approximately a 20% lower instantaneous rate of reassessment throughout the ED stay. Medicare was also associated with slower reassessment (HR 0.886 (95% CI 0.838–0.936), $p < 0.001$), though the magnitude was smaller than for Medicaid.

In contrast to the race/ethnicity findings, the Medicaid association showed minimal attenuation across sequential models: from M2 (0.826 (95% CI 0.782–0.872), $p < 0.001$) through M3 (0.820 (95% CI 0.776–0.866), $p < 0.001$) to M4 (0.807 (95% CI 0.763–0.853), $p < 0.001$). This stability indicates that the insurance disparity is not primarily explained by differences in triage acuity, clinical severity, arrival mode, or shift distribution between Medicaid and privately insured patients. The insurance effect appears to operate through channels not captured by the measured covariates in this model (see Figure 1, dashed insurance direct-effect arrow).

Within-acuity analyses corroborated this pattern (Figure 7). Among ESI 1–2 encounters, Medicaid patients were reassessed more slowly than privately insured patients (HR 0.752 (95% CI 0.685–0.825), $p < 0.001$). Among ESI 3 encounters, Medicaid remained associated with slower re-

assessment (HR 0.838 (95% CI 0.776–0.904), $p < 0.001$). Estimates in ESI 4–5 were imprecise due to smaller event counts. The persistence of the insurance association even among the highest-acuity patients argues against the explanation that Medicaid patients are simply lower-acuity on average.

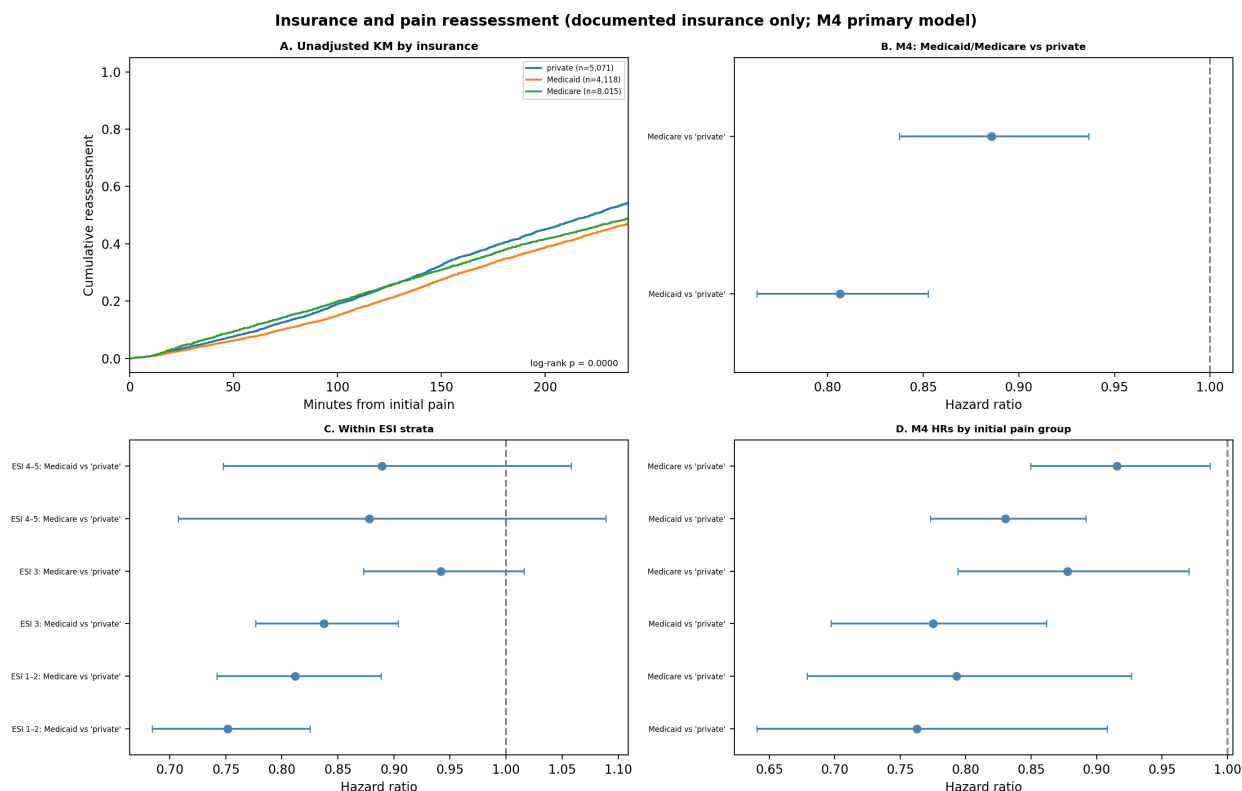


Figure 7: Insurance-focused analysis: unadjusted curves, M4 hazard ratios, within-acuity Medicaid estimates, and pain-stratum models.

3.6 Severe-pain and within-acuity sensitivity analyses

Among patients reporting severe pain (initial pain score 7–10), an insurance-based disparity in reassessment persisted (HR 0.837 (95% CI 0.736–0.950), $p = 0.006$). These are patients whose self-reported symptom burden should, if the system were responding to the clinical signal alone, uniformly trigger monitoring. The fact that the insurance gradient survives in this subgroup is among the most important findings of this analysis: it cannot be attributed to Medicaid patients reporting lower pain or being perceived as lower-acuity, because this restriction excludes those patients entirely.

Within-acuity models showed a similar pattern (Figure 8). The Black–White difference in the severe-pain subgroup was HR 0.854 (95% CI 0.763–0.956), $p = 0.006$, consistent with the primary analysis and subject to the same triage-pathway interpretation described in Section 3.4.

Within-acuity M4-style models (race/insurance; no disposition or analgesia)

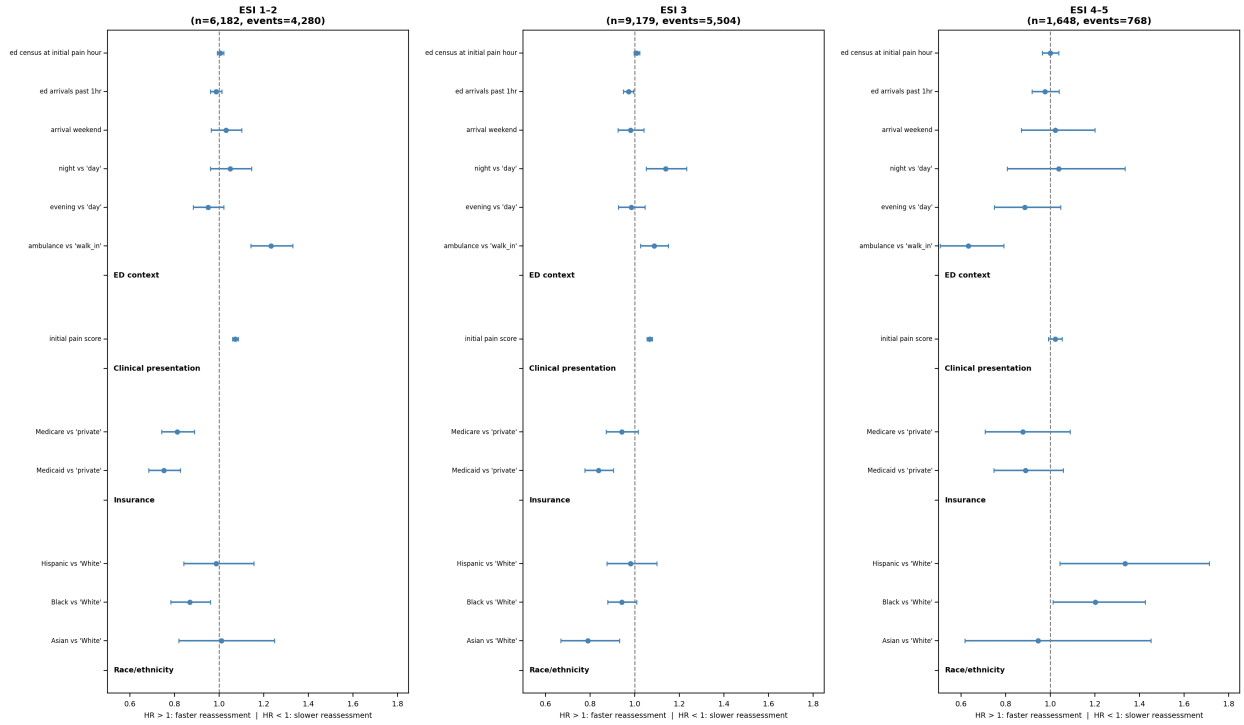


Figure 8: Within-acuity M4-style models: race/ethnicity and insurance by ESI stratum.

3.7 Temporal trends

There was no evidence of a meaningful change in reassessment rates over the study period. In a supplementary Cox model with continuous de-identified year, the HR per 5-year increase was 0.999 (95% CI 0.996–1.003, $p = 0.793$). Figure 10 displays unadjusted and M4-adjusted probabilities of 60-minute reassessment by policy era; both series fluctuated between approximately 7% and 12% without a consistent directional pattern, indicating that reassessment practices did not improve over the observation window despite national emphasis on pain management quality during this interval.

3.8 Post-analgesic pathway analysis

When time zero was reset to the first analgesic administration, cumulative reassessment curves by race/ethnicity largely overlapped and the multivariate log-rank test was not statistically significant in the primary trauma severe-pain subgroup (Figure 11). This finding indicates that the demographic patterns observed in the primary analysis — particularly the race/ethnicity associations — are not primarily driven by differences in how quickly patients are monitored after receiving treatment. Rather, the earlier-encounter phase (between initial pain documentation and first reassessment, irrespective of treatment) appears to be where differences in clinical attention emerge.

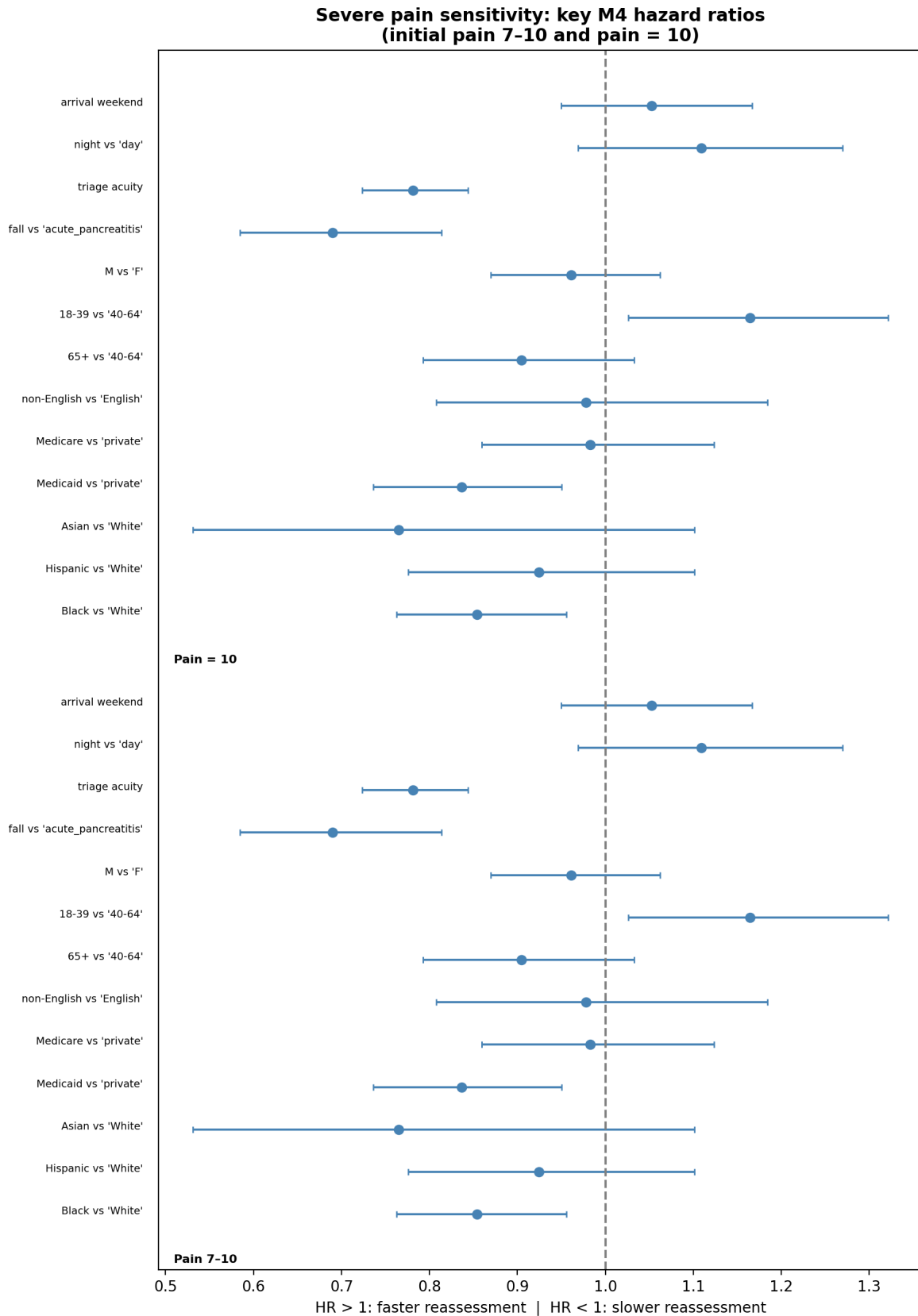


Figure 9: Severe-pain sensitivity analysis (pain 7–10): key M4 hazard ratios.

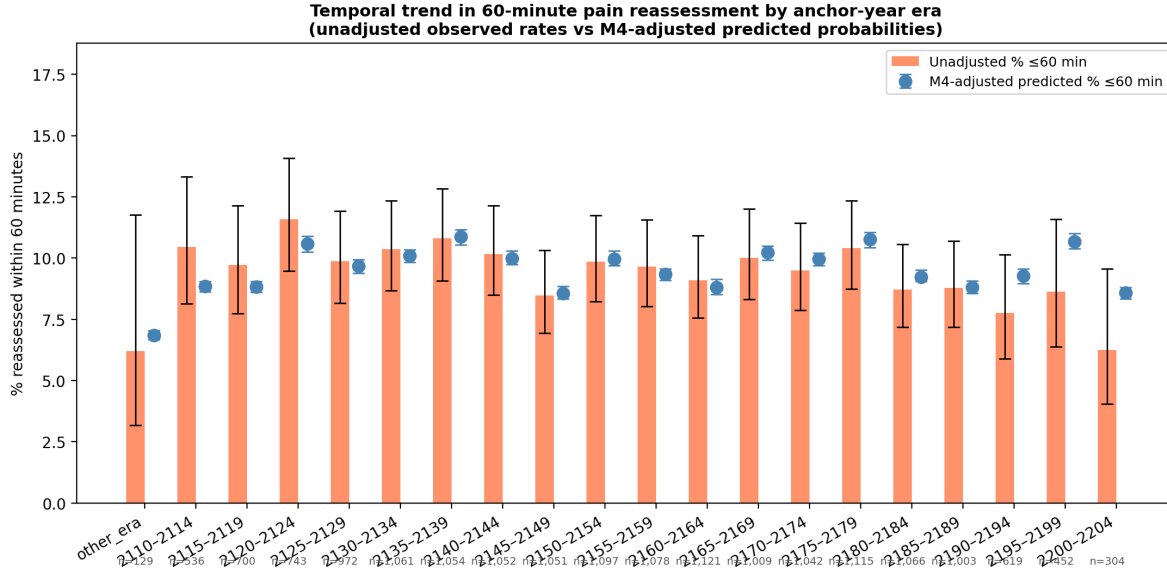


Figure 10: Year-over-year trends in 60-minute reassessment by de-identified policy era: unadjusted observed rates (bars) and M4-adjusted predicted probabilities (points) with 95% confidence intervals.

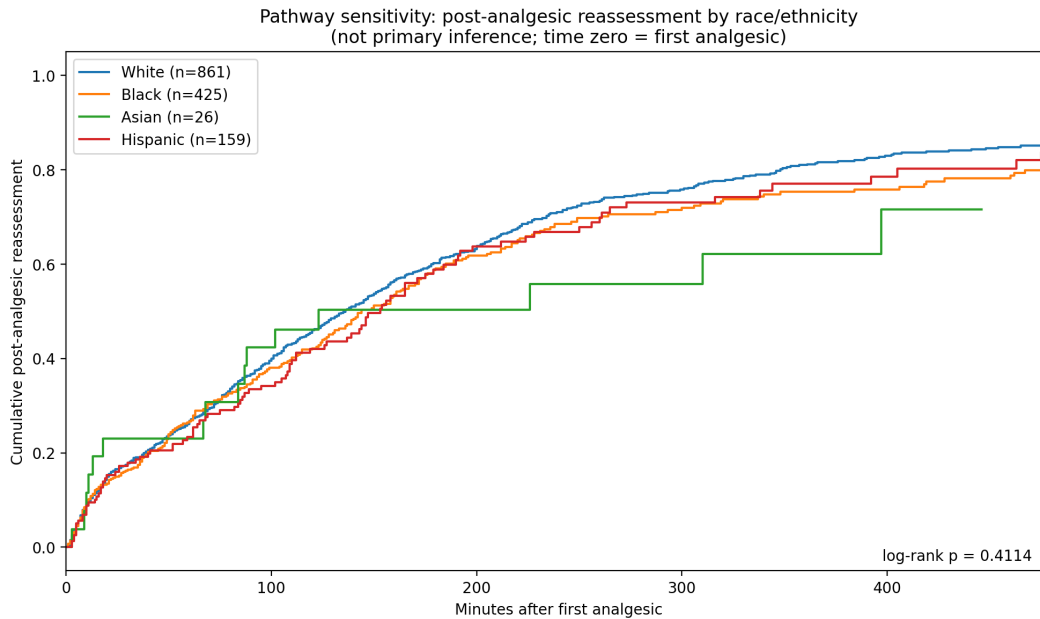


Figure 11: Post-analgesic pathway sensitivity: reassessment by race/ethnicity with time zero at first analgesic administration.

4 Discussion

In this retrospective cohort of ED encounters with documented pain assessments, we asked whether the emergency department responds equally to a patient’s reported pain — and found that it does not. Reassessment was frequently delayed: fewer than one in ten patients had their pain reassessed within 60 minutes of initial documentation, and the median time to reassessment exceeded three hours. The factors associated with reassessment timing form a coherent picture: clinical signals (higher reported pain, ambulance arrival) accelerate the system’s response, while operational and social factors introduce substantial variation beyond what clinical severity alone would predict.

4.1 The clinical response to pain is appropriately shaped by severity — but incompletely

The strongest and most clinically expected associations were with the pain report itself and with triage acuity. Patients who reported higher pain were reassessed more quickly; patients assigned lower ESI acuity were reassessed more slowly. This pattern is broadly consistent with appropriate clinical prioritization: the emergency department allocates monitoring attention in proportion to perceived urgency. Night-shift associations likely reflect lower census conditions that facilitate more frequent patient contact.

However, the aggregate reassessment rates in this cohort suggest that even among patients with high reported pain, timely reassessment is not the norm. Only 9.6% of patients were reassessed within 60 minutes. If the system is responding appropriately to a pain signal, this baseline rate warrants attention as a quality benchmark independent of any disparity analysis.

4.2 ESI attenuation as evidence of a disparity mechanism, not its absence

The most important interpretive point in this paper concerns what the attenuation of race and ethnicity coefficients across sequential models actually means. Black patients had slower reassessment than White patients in models adjusted only for clinical presentation and demographics. This association attenuated substantially after adding ESI triage acuity. A naive reading of this pattern — common in epidemiologic reporting — would conclude that the disparity disappears once clinical factors are controlled, implying no inequity once case mix is accounted for. That reading is incorrect, and stating it plainly matters for how findings like these get translated into policy.

ESI triage acuity is not a neutral measure of objective clinical severity. It is a judgment, made by a provider, about how urgently a patient’s presentation requires attention. That judgment is susceptible to the same implicit biases documented in pain treatment, analgesic prescribing, and clinical assessment more broadly. Empirical evidence supports this: studies have found that Black patients are more likely to be triaged to lower-acuity or fast-track areas relative to White patients with equivalent presenting complaints [Vigil et al., 2014, Pines et al., 2009], and that the ESI instrument relies on subjective determinations — particularly resource prediction — that are especially prone to unintentional racial bias [Fernandes et al., 2005]. When a Black patient and a

White patient present with the same pain score and the same diagnosis, and receive different ESI assignments, those assignments carry the disparity forward into every downstream outcome that ESI predicts — including reassessment timing.

Adjusting for ESI in this setting does not control for confounding; it blocks a causal pathway. The attenuation of the Black–White coefficient from M2 to M3 is therefore evidence that differential triage assignment is a *mechanism* through which racial disparities in reassessment arise — not evidence that the disparity is explained away. The residual M4 coefficient represents the portion of the disparity not mediated through measured triage and workflow variables, not the total disparity. Total disparity estimates that do not adjust for ESI, or that treat ESI as a mediator rather than a confounder, would show larger and more persistent race effects.

This distinction matters for intervention design. If racial disparities in reassessment operate through triage assignment, then interventions that target downstream monitoring protocols without addressing triage equity will be insufficient. The triage process itself — the point at which a provider encodes their assessment of urgency — is where the disparity is generated.

4.3 Insurance: a disparity not explained by the care pathway

The most striking evidence of inequitable pain monitoring in this study is not the average Medicaid–private difference across the full cohort — it is the persistence of that difference among patients reporting severe pain. Among patients with initial pain scores of 7–10, Medicaid patients were still reassessed substantially more slowly than privately insured patients (HR 0.837 (95% CI 0.736–0.950), $p = 0.006$). These patients have already cleared the threshold that should, on clinical grounds alone, prompt monitoring: they reported severe pain at triage. The insurance gradient that persists in this subgroup is therefore not attributable to Medicaid patients appearing lower-acuity, reporting lower pain, or being clinically less urgent. It reflects something else — something in the care encounter itself that is associated with payer status and that shapes how often patients are checked on.

Across the full cohort, the Medicaid association showed minimal attenuation from M2 through M4, in contrast to the race and ethnicity findings. Medicaid patients were reassessed approximately 20% more slowly than privately insured patients at the same triage level, with the same initial pain score, arriving by the same mode, during the same shift. Within-acuity analyses confirm this pattern persists even among ESI 1–2 patients (HR 0.752 (95% CI 0.685–0.825), $p < 0.001$) — those whose triage designation explicitly signals the highest clinical urgency. When the clinical system has already correctly identified a patient as high-priority, insurance status should have no bearing on how frequently that patient is monitored. That it does suggests the disparity is embedded in the routine fabric of the care encounter rather than in misclassification of clinical urgency.

The unmeasured mechanisms are plausible: differences in how nursing attention is distributed across patients, in how assertively patients advocate for themselves, in implicit provider assumptions about which patients require close monitoring, or in structural factors such as bed assignment and staffing ratios that correlate with payer mix at the unit level. Distinguishing among these

mechanisms requires prospective data beyond what administrative records can provide.

What administrative data can establish — and what this study does establish — is that the disparity is real, robust, and not attributable to measured clinical or operational factors. Because reassessment is directly observable, protocol-adjacent, and not dependent on a subjective clinical judgment, it is a more tractable quality-improvement target than treatment disparities. Structured reassessment protocols tied to ESI level, nursing workflow checklists, or electronic alerts for time-since-last-pain-documentation are all interventions that operate at the level where this disparity appears to live.

4.4 Temporal stability

Reassessment rates did not improve meaningfully across the study period, despite national emphasis on pain management quality during this interval. Both unadjusted and adjusted estimates of 60-minute reassessment fluctuated between approximately 7% and 12% without directional improvement across policy eras. This stability suggests that reassessment documentation practices were not materially affected by broader quality initiatives — motivating explicit inclusion of reassessment metrics in future quality measurement frameworks.

4.5 Limitations

This study has several important limitations. Data come from a single academic medical center via MIMIC-IV; findings may not generalize to community or safety-net EDs, which may have different staffing structures, patient populations, and documentation practices. The cohort was restricted to acute pancreatitis and trauma diagnoses; findings should not be extrapolated to all ED pain presentations. Pain scores and reassessment timestamps reflect documentation behavior, which may differ from the underlying clinical activity — a nurse who checks on a patient but does not document a pain score would not contribute a reassessment event in this analysis. We could not fully account for analgesic timing, provider-level variation, or patient advocacy behaviors that may mediate or moderate the associations we observe. Comorbidity measurement depended on linked ICD flags, which may be incomplete. De-identified calendar years prevent direct mapping to specific policy interventions or institutional changes. Finally, causal interpretation of the sequential model coefficients requires the DAG assumptions to hold; if ESI assignment is influenced by unmeasured variables that also affect reassessment, the structural interpretation of the attenuation pattern may be incorrect.

4.6 Conclusions

Pain is a subjective experience that requires documentation to become visible to the clinical system, and reassessment to be managed over time. In this cohort, the emergency department’s response to a patient’s reported pain varied substantially with clinical, operational, and social factors — and this variation was not fully explained by differences in measured clinical severity. The pattern of

results suggests that inequities in pain reassessment may be embedded in triage pathways for some demographic groups and in unmeasured aspects of care allocation for others. Pain reassessment should be considered alongside analgesic outcomes in equity-oriented quality assessment. Future work should validate these associations in multi-center cohorts, link reassessment timing to patient-reported pain outcomes, and evaluate targeted interventions to standardize reassessment across payer groups and triage strata.

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Supplementary Materials

All supplementary figures are generated by the analysis pipeline in `figures/manuscript/appendix/`. Code to reproduce all analyses and figures is available in `scripts/run_analysis.py`.

Table S1: **Vital-sign hazard ratios from the M4 model.** Initial heart rate, respiratory rate, and systolic blood pressure (each z-scored) are adjusted for in M3 and M4 but are not displayed in the primary forest plot (Figure 5 in main text). HR per one standard-deviation increase in each vital sign.

Vital sign (z-scored)	HR (95% CI)	p-value
Heart rate	1.025 (1.004–1.047)	0.019
Respiratory rate	1.009 (0.999–1.019)	0.082
Systolic blood pressure	0.995 (0.977–1.013)	0.575

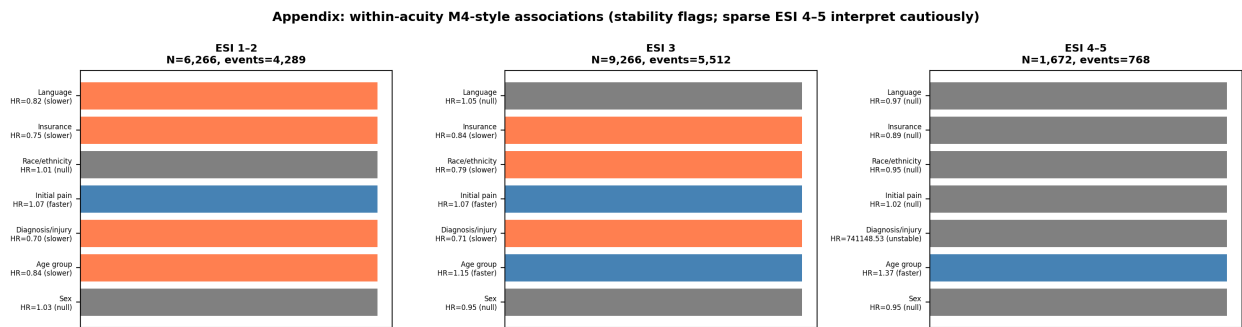


Figure S1: **Appendix: within-acuity M4-style associations (stability flags; sparse ESI 4–5 interpret cautiously).** Estimated associations within ESI strata for age, sex, diagnosis, and language; complements main-text within-acuity results for insurance and race/ethnicity (Figure 8).

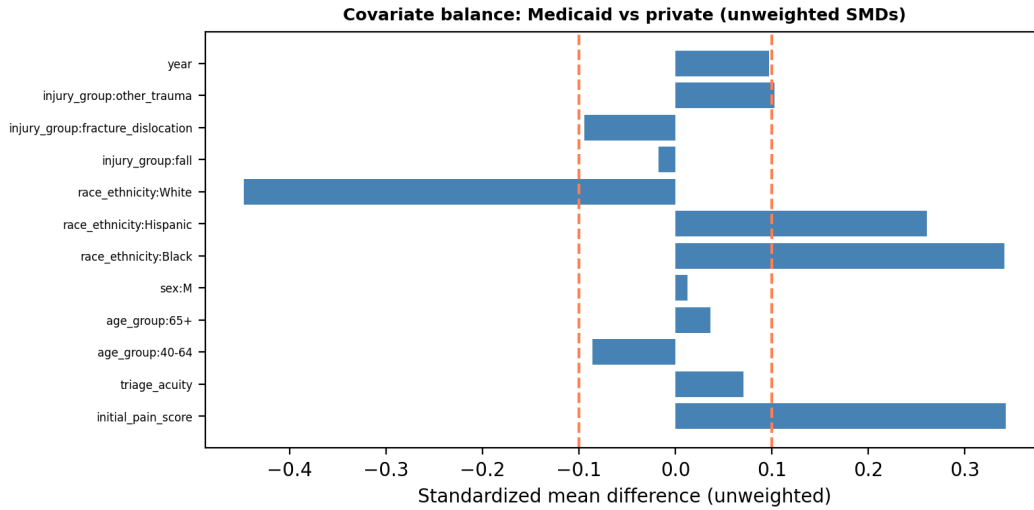


Figure S2: **Covariate balance: Medicaid vs. private (unweighted SMDs)**. Standardized mean differences before inverse probability of treatment weighting (IPTW), using propensity scores estimated from M1–M3 covariates.

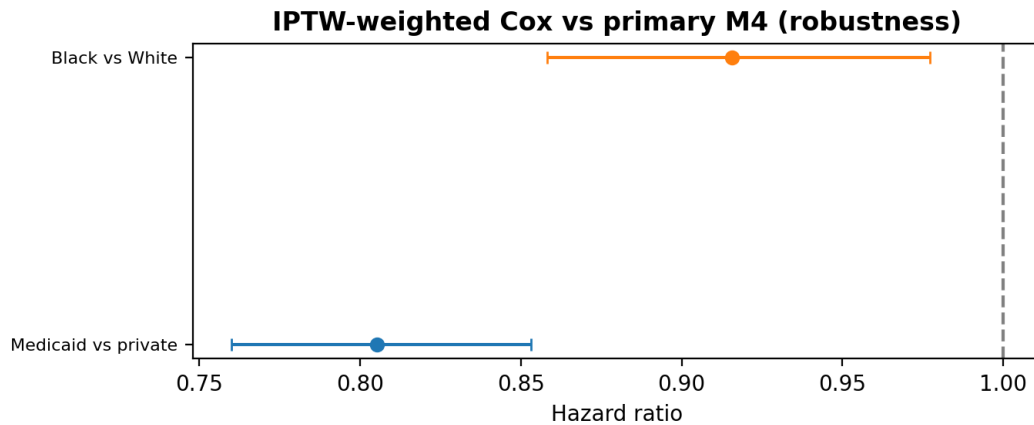


Figure S3: **IPTW-weighted Cox vs. primary M4 (robustness)**. Weighted hazard ratio for Medicaid vs. private insurance compared with the unweighted M4 estimate.

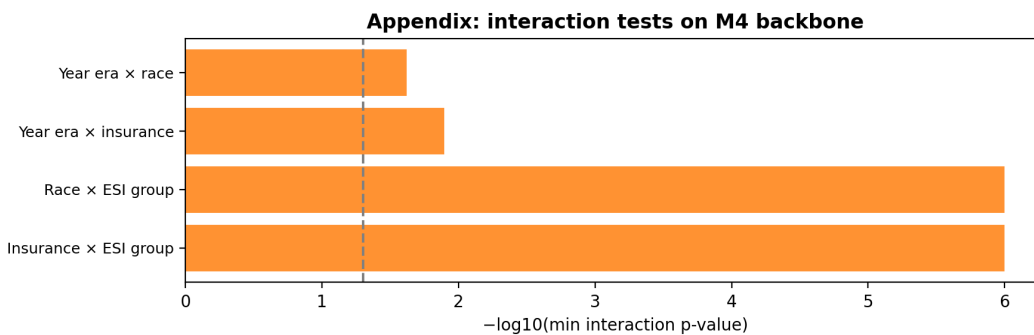


Figure S4: **Appendix: interaction tests on M4 backbone**. Race $\times$ ESI, insurance $\times$ ESI, and year-era interaction terms on the M4 covariate set.